

Validity Analysis: MASSIVE

Target context: Ethiopian Intercity Travelers — Amharic-Language Booking Bot

Overall risk: **HIGH**

Deployment Context

Use case: Amharic intercity-travel booking bots for Sky Bus, Selam Bus, and Ride covering routes, fares, and seat changes

Target population: Inter-regional commuters and travelers booking long-distance buses and ride-hail trips via smartphone apps

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology	2	Significant gaps	medium
Input Content $\triangle$	1	Serious concern	high
Input Form $\checkmark$	4	Minor gaps	medium
Output Ontology	2	Significant gaps	medium
Output Content $\triangle$	1	Serious concern	high
Output Form $\checkmark$	4	Minor gaps	high
Average	2.3		

$\triangle$ = highest concern $\checkmark$ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MASSIVE provides a strong modality and output-format match for an Amharic intent-classification + slot-filling booking bot, but exhibits severe content-level gaps for the Ethiopian intercity-travel deployment. The translation-from-English pipeline [Q2, Q4, Q106] guarantees that Amharic data lacks Ethiopian calendar dates, local city name variants, Birr fare conventions, and code-switching with Oromo or Tigrinya — all of which the user has confirmed appear in real user utterances. Operator-specific intents and slot types (operator_name, seat_class, route_closure_reason, surcharge_event, departure_terminal) are absent from the 60-intent / 55-slot taxonomy, which the paper itself describes as 'relatively simple' [Q108]. Annotator demographics for Amharic are undisclosed, raising convergent-validity concerns about whether labels reflect Ethiopian intercity-traveler judgments. Per-language data is small (~19.5k records, [Q103]) and Amharic — a unique-script, lower-resource language — is

expected to fall in the lower performance tier per the script/pretraining-data correlation [Q87, Q98, Q100]. Strengths: text-only modality match, Ethiopic-script inclusion, output format directly compatible, and rich per-language breakdowns enabling expected-performance estimation.

ID	Evidence
Q2	"MASSIVE was created by tasking professional translators to localize the English-only SLURP dataset into 50 typologically diverse languages from 29 genera." (p.1)
Q4	"MASSIVE was created by localizing the SLURP NLU dataset (created only in English) in a parallel manner." (p.1)
Q87	"We compared the pretraining data quantities by language for XLM-R to its per-language task performance values, and in the zero shot setup, we found a Pearson correlation of 0.54 for exact match" (p.7)
Q98	"Discounting for artificial spacing effects, Germanic genera and Latin scripts performed the best overall (See Appendix E), which is unsurprising given the amount of pretraining data for those genera and scripts, as well as the quantity of Germanic and Latin-script languages in MASSIVE." (p.8)
Q100	"Icelandic was the lowest-performing Germanic language, likely due to a lack of pretraining data, as well as to its linguistic evolution away from the others due to isolated conditions." (p.8)
Q103	"Starting with the dataset, the per-language data quantities are relatively small at 19.5k total records and 11.5k records for training." (p.9)
Q106	"Third, the data were originally created through crowd-sourcing, not from a real virtual assistant, which introduces artificialities." (p.9)
Q108	"Fourth, our labeling schema is relatively simple when compared with hierarchical labeling schemata or flat schemata with more intent and slot options." (p.9)

Practical Guidance

What This Benchmark Measures

MASSIVE provides a baseline for Amharic intent-classification and slot-filling on translated-from-English virtual-assistant utterances [Q2, Q5, Q73]. For the deployment, it can establish a lower-bound expectation for how XLM-R or mT5-class models handle generic booking-style intents in Amharic and provide per-language performance figures [Q126, Q127, Q128] useful for capacity planning. Output Form (score 4) and Input Form (score 4) are the strongest dimensions and indicate that the evaluation paradigm itself transfers cleanly.

Construct Depth

Construct depth is shallow for the deployment. MASSIVE measures translation-quality intent/slot accuracy on a generic taxonomy, not naturalistic Ethiopian intercity-booking utterances. Input Content (score 1) and Output Content (score 1) gaps mean the benchmark cannot probe Ethiopian-calendar slot extraction, Birr fare normalization, city-name-variant handling, code-switching, or L2-speaker phrasing — the phenomena most likely to drive real-world failure. Input Ontology (score 2) and Output Ontology (score 2) gaps mean operator-specific intents and slot types are entirely outside the evaluation surface.

ID	Evidence
Q2	"MASSIVE was created by tasking professional translators to localize the English-only SLURP dataset into 50 typologically diverse languages from 29 genera." (p.1)
Q5	"We also present modeling results on XLM-R and mT5, including exact match accuracy, intent classification accuracy, and slot-filling F1 score." (p.1)
Q73	"The decoder output is a sequence of labels (including the Other label) for all of the tokens followed by the intent." (p.7)
Q126	"Exact match accuracy by language for our three models using the full dataset and the zero-shot setup." (p.20)
Q127	"Table 8: Intent accuracy by language for our three models using the full dataset and the zero-shot setup." (p.21)
Q128	"Micro-averaged slot-filling F1 by language for our three models using the full dataset and the zero-shot setup." (p.22)

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: medium

Benchmark: MASSIVE | Context: Ethiopian Intercity Travelers — Amharic-Language Booking Bot

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MASSIVE's taxonomy of 18 domains, 60 intents, and 55 slots **[Q1]** was designed around major virtual assistants **[Q16]** and the 'general multi-domain NLU task' **[Q12]**. The corpus consists primarily of interrogatives and imperatives **[Q30]**, which broadly aligns with booking-bot utterances. However, the taxonomy contains no intercity-transport-specific intents (route_query, operator_query, seat_class_query, cancel_and_rebook for road closures, holiday_surcharge_query, departure_terminal_query) that are explicitly required for the deployment, and the paper acknowledges its labeling schema is 'relatively simple when compared with hierarchical labeling schemata or flat schemata with more intent and slot options' **[Q108]**. The user views booking flows as broadly generic (Q1 elicitation), reducing concern, but at least six intent types relevant to the deployment are absent, and a confirmed full gap exists across African NLP for transport-domain NLU intents (search_target gap_id 6).

ID	Evidence
Q1	"MASSIVE contains 1M realistic, parallel, labeled virtual assistant utterances spanning 51 languages, 18 domains, 60 intents, and 55 slots." (p.1)
Q12	"This work focuses on the general multi-domain NLU task and builds off the SLURP (Bastianelli et al., 2020) benchmark dataset to extend to an unprecedented 50 new languages." (p.2)
Q16	"Second, we determined existing languages available in major virtual assistants, such that the dataset could be used to benchmark today's systems." (p.3)
Q30	"Specifically, the corpus primarily consists of interrogatives (i.e., questions) and imperatives (commands or requests)." (p.4)
Q108	"Fourth, our labeling schema is relatively simple when compared with hierarchical labeling schemata or flat schemata with more intent and slot options." (p.9)

Strengths

- Question and command coverage matches booking-bot interaction patterns — corpus 'primarily consists of interrogatives (i.e., questions) and imperatives (commands or requests)' **[Q30]**
- Includes Amharic as one of 51 languages, with WALS-derived typological metadata that lets evaluators stratify performance by SOV order, verb morphology, etc. **[Q34, Q37]**

ID	Evidence
Q30	"Specifically, the corpus primarily consists of interrogatives (i.e., questions) and imperatives (commands or requests)." (p.4)
Q34	"In MASSIVE, 39 languages are subject-initial (24 SVO and 15 SOV), while only three are verb-initial (VSO specifically)." (p.4)
Q37	"The majority of them (33) use some kind of verb morphology, such as adding a suffix." (p.4)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories

IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

- IO-1:** Required deployment categories include `route_query`, `operator_query`, `seat_class_query`, `cancel_and_rebook`, `holiday_surcharge_query`, `departure_terminal_query` (regional context, `booking_flow_intents.absent_intents`). Generic booking intents (`book`, `reschedule`, `check_availability`) are partially covered.
- IO-2:** MASSIVE's 60-intent taxonomy was selected to 'benchmark today's [virtual assistant] systems' [Q16] and omits intercity-transport-specific intents listed above. Confirmed by gap_id 6 search [WEB-21, WEB-22] showing no African transport NLU benchmark exists.
- IO-3:** MASSIVE includes domains (smart-home, music, IoT-style commands implied by SLURP heritage [Q12, Q13]) that are irrelevant to a transport booking deployment, contributing construct-irrelevant variance to overall accuracy metrics. INSUFFICIENT DOCUMENTATION on per-domain relevance scoring; would need a domain-by-domain mapping to deployment scope.
- IO-4:** Significant content-validity gaps: Ethiopian-calendar holiday surcharge intents, route-closure rebooking, operator/seat-class queries, and terminal-station queries are absent [Q108; gap_id 6]. The booking flow's transactional core (`book/cancel/reschedule`) is partially captured but operator-specific scenarios are not.

ID	Evidence
Q1	"MASSIVE contains 1M realistic, parallel, labeled virtual assistant utterances spanning 51 languages, 18 domains, 60 intents, and 55 slots." (p.1)
Q12	"This work focuses on the general multi-domain NLU task and builds off the SLURP (Bastianelli et al., 2020) benchmark dataset to extend to an unprecedented 50 new languages." (p.2)
Q13	"The Snips datasets (both the original English only and the English and French releases) are most similar to the NLU contained in the MASSIVE dataset, spanning smart home and music domains for a generic voice-based virtual assistant." (p.2)
Q16	"Second, we determined existing languages available in major virtual assistants, such that the dataset could be used to benchmark today's systems." (p.3)
Q30	"Specifically, the corpus primarily consists of interrogatives (i.e., questions) and imperatives (commands or requests)." (p.4)
Q108	"Fourth, our labeling schema is relatively simple when compared with hierarchical labeling schemata or flat schemata with more intent and slot options." (p.9)
WEB-21	No African transport-domain slot taxonomy exists — operator-specific slot types must be developed locally (https://aclanthology.org/2024.lrec-main.561/)
WEB-22	EthioLLM/EthioBenchmark arXiv — no booking-domain NLU resource (https://arxiv.org/abs/2403.13737)

Information Gaps

- Per-domain mapping showing what fraction of MASSIVE's 18 domains/60 intents are relevant to intercity bus booking
- Complete operator-specific intent list from Sky Bus, Selam Bus, and Ride

Requires Expert Verification

- Whether booking-flow elicitation in Ethiopia would surface additional intent types beyond the listed gaps
- Whether holiday-surcharge or road-closure intents are framed by users as separate intents or as parameter values within general booking intents

Remediation

Gap 1: Intercity-transport-specific intents (`route_query`, `operator_query`, `seat_class_query`, `cancel_and_rebook`, `holiday_surcharge_query`, `departure_terminal_query`) absent from MASSIVE's 60-intent taxonomy

Recommendation: Extend the intent taxonomy with at least these six transport-specific intents through stakeholder elicitation with Sky Bus, Selam Bus, and Ride operators (gap_id 11). Ensure each new intent has ≥ 100 elicited Amharic examples for training and ≥ 50 for held-out testing.

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MASSIVE | Context: Ethiopian Intercity Travelers — Amharic-Language Booking Bot

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MASSIVE's Amharic content was created by 'tasking professional translators to localize the English-only SLURP dataset' [Q2, Q4], with seed data 'originally created through crowd-sourcing, not from a real virtual assistant, which introduces artificialities' [Q106]. This means Amharic instances will not contain Ethiopian calendar dates (Meskerem, Tikimt — 13-month Ge'ez calendar offset 7–8 years from Gregorian, [WEB-4]), local city name variants (Adama/Nazret, Mekelle/Mek'ele), Birr-denominated fares, or holiday-surcharge contexts (Timkat, Meskel, Enkutatash — [WEB-5, WEB-6]) — all of which the user has confirmed are required slot values (Q2 elicitation). Worker-decided localization 'added further noise' [Q107], and per-language data is 'relatively small at 19.5k total records and 11.5k records for training' [Q103]. The IC dimension was rated HIGH priority and the mismatch is severe.

ID	Evidence
Q2	"MASSIVE was created by tasking professional translators to localize the English-only SLURP dataset into 50 typologically diverse languages from 29 genera." (p.1)
Q4	"MASSIVE was created by localizing the SLURP NLU dataset (created only in English) in a parallel manner." (p.1)
Q103	"Starting with the dataset, the per-language data quantities are relatively small at 19.5k total records and 11.5k records for training." (p.9)
Q106	"Third, the data were originally created through crowd-sourcing, not from a real virtual assistant, which introduces artificialities." (p.9)
Q107	"Relatedly, allowing the worker to decide on translation versus localization of slot entities added further noise to the dataset, although we try to store this decision in the metadata." (p.9)
WEB-4	Ethiopian (Ge'ez) calendar: 13 months, 7–8-year offset from Gregorian — temporal expression slot values absent from MASSIVE (https://ethiocal.com/)
WEB-5	Timkat/Ethiopian Epiphany — major travel surge, Tir 11 E.C. (https://www.ethiopiancalendar.net/ethiopian-holidays)
WEB-6	Meskel, Genna, Enkutatash — peak travel periods unrepresented in MASSIVE (https://holidays.ertale.com/)

Strengths

- Translation workflow allows slot localization — workers could localize values like song names [Q57, Q58], so some Amharic content adaptation occurred
- Metadata stores the translate/localize/unchanged decision per slot [Q58], enabling researchers to filter for genuinely localized content

ID	Evidence
Q57	"The worker is asked to either localize or translate the slot, depending on whether the value should be translated (e.g., "tomorrow") or localized (e.g., the movie "La La Land", which in French is "Pour l'amour d'Hollywood.")" (p.6)
Q58	"The metadata of the released dataset includes whether the worker elected to "localize," "translate," or keep the slot "unchanged," primarily for the purposes of researchers evaluating machine translation systems, where it would be unreasonable to expect the system to "localize" to a specific song name the worker selected." (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — the deployment requires Ethiopian calendar dates (Meskerem, Tikimt — [\[WEB-4\]](#)), local city names with Amharic spellings and alternate variants, Birr fare conventions, and operator/route knowledge specific to Sky Bus, Selam Bus, Ride. None of these are present in MASSIVE's English-derived content [\[Q2, Q4, Q106\]](#).

IC-2: MASSIVE's content originates from English crowd-workers [\[Q106\]](#) and was translated into Amharic [\[Q2\]](#); cultural alignment with Ethiopian intercity travelers is incidental. Holiday concepts, calendar conventions, and currency expressions are not Ethiopianized.

IC-3: Yes — the seed SLURP data reflects Western/Anglophone virtual-assistant interaction patterns (smart-home, music, weather domains [\[Q9, Q13, Q14\]](#)). Localization noise [\[Q107\]](#) further means inputs may carry residual Western reference frames.

IC-4: Recruitment of Ethiopian intercity-traveler-representative annotators is necessary; gap_id 5 confirms no L2-Amharic-speaker NLP dataset exists, and gap_id 11 confirms operator-specific data requires direct elicitation. INSUFFICIENT DOCUMENTATION on Amharic translator demographics in MASSIVE.

IC-5: Severe content-validity issues: absence of Ethiopian temporal expressions, locally varied city names, Birr fares, and code-switching with Oromo/Tigrinya (gaps 1, 2, 3, 4, 7) means the benchmark's Amharic inputs do not represent the target distribution. This is the dominant validity concern.

ID	Evidence
Q2	"MASSIVE was created by tasking professional translators to localize the English-only SLURP dataset into 50 typologically diverse languages from 29 genera." (p.1)
Q4	"MASSIVE was created by localizing the SLURP NLU dataset (created only in English) in a parallel manner." (p.1)
Q9	"The first iteration for the foundation of the MASSIVE dataset was the NLU Evaluation Benchmarking Dataset, with 25k utterances across 18 domains (Liu et al., 2019a)." (p.2)
Q13	"The Snips datasets (both the original English only and the English and French releases) are most similar to the NLU contained in the MASSIVE dataset, spanning smart home and music domains for a generic voice-based virtual assistant." (p.2)
Q14	"Facebook released a general Intelligent Virtual Assistant (IVA) dataset across the domains of Alarm, Reminder, and Weather (Schuster et al., 2019) created for the purpose of demonstrating cross-lingual transfer learning; and so did not need to be parallel or have an equal number of datapoints, resulting in far fewer examples in Thai (5k) compared to Spanish (7.6k) and English (43k)." (p.2)
Q103	"Starting with the dataset, the per-language data quantities are relatively small at 19.5k total records and 11.5k records for training." (p.9)
Q106	"Third, the data were originally created through crowd-sourcing, not from a real virtual assistant, which introduces artificialities." (p.9)
Q107	"Relatedly, allowing the worker to decide on translation versus localization of slot entities added further noise to the dataset, although we try to store this decision in the metadata." (p.9)
WEB-4	Ethiopian (Ge'ez) calendar: 13 months, 7–8-year offset from Gregorian — temporal expression slot values absent from MASSIVE (https://ethiocal.com/)
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WEB-6	Meskel, Genna, Enkutatash — peak travel periods unrepresented in MASSIVE (https://holidays.ertale.com/)
WEB-17	amseg Amharic segmenter/tokenizer (Yimam et al. 2021) exists (https://github.com/EthioNLP/Ethiopian-Language-Survey)
WEB-21	No African transport-domain slot taxonomy exists — operator-specific slot types must be developed locally (https://aclanthology.org/2024.lrec-main.561/)

Information Gaps

- What fraction of Amharic slot instances in MASSIVE were marked 'localize' vs. 'translate' vs. 'unchanged' [Q58 metadata not summarized in paper]
- Whether any Amharic city names (e.g., Addis Ababa, Bahir Dar) appear in MASSIVE slot values

Requires Expert Verification

- Confirmation that Ethiopian intercity travelers' real utterances frequently mix Birr/Ethiopian-calendar/local-variant values with code-switching
 - Whether MASSIVE's Amharic data (despite limitations) provides sufficient typological prior for transfer learning to a finetuned booking model
-

Remediation

Gap 1: Ethiopian-calendar dates, local city name variants (Adama/Nazret, Mekelle/Mek'ele), Birr fare expressions, and holiday-surcharge contexts are absent from translated MASSIVE Amharic data

Recommendation: Construct a supplementary slot-filling test set of 1–3k Amharic booking utterances elicited from Ethiopian intercity travelers, covering Ge'ez calendar months, alternate city names, Birr numerals (Arabic, Ethiopic, and verbalized forms), and named holidays (Timkat, Meskel, Enkutatash, Genna, Fasika, Eid). Use this as the primary acceptance test, with MASSIVE Amharic as a secondary baseline.

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: medium

Benchmark: MASSIVE | Context: Ethiopian Intercity Travelers — Amharic-Language Booking Bot

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

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Justification

Modality alignment is strong: MASSIVE is text-only [\[Q3\]](#) and Amharic is included as one of 51 languages, with 21 distinct scripts represented [\[Q25\]](#) including Ethiopic/Ge'ez fidel as a 'unique' script [\[Q27\]](#). The deployment is also text-only smartphone input in Ethiopic script. Script diversity was explicitly sought 'to drive experimentation in tokenization and normalization' [\[Q20\]](#). The paper acknowledges 'our collection system did not have a robust method to preserving or denoting native tokenization practices' [\[Q109\]](#), leaving slot-boundary handling for Amharic's morphologically rich, fusional structure as an open question (gap_id 8 PARTIAL — amseg and HornMorpho exist but not evaluated against MASSIVE Amharic). Amharic uses spaces as word delimiters, so unlike Japanese/Chinese/Khmer it does not require character-level splitting [\[Q80, Q92, Q96\]](#). The IF dimension is rated LOWER priority and modality match is good, but documentation of Amharic-specific tokenization quality is absent.

ID	Evidence
Q3	"With the English seed data included, there are 587k train utterances, 104k dev utterances, 152k test utterances, and 153k utterances currently held out for the MMNLU-22 competition, which will be released after the competition." (p.1)
Q20	"Fifth, we examined the script of each language, seeking to increase script diversity to drive experimentation in tokenization and normalization." (p.3)
Q25	"There are 21 distinct scripts used in the dataset." (p.3)
Q27	"The Arabic script is used for three languages, the Cyrillic script for two languages, and the remaining 18 languages have "unique" scripts, in the sense that only one language in the dataset uses that script." (p.3)
Q80	"Three of these written languages, Japanese, Chinese (Traditional), and Chinese (Simplified), do not use spaces anywhere except to identify the end of a sentence. For these languages, we separate each character in the unlabeled input with a whitespace." (p.7)
Q92	"For Khmer, the workers had a difficult time adapting their translations and localizations to properly-slotted outputs given the space-optional nature of the language." (p.8)
Q96	"character spacing was necessary in order to properly assign the slots to the right characters." (p.8)
Q109	"Fifth, our collection system did not have a robust method to preserving or denoting native tokenization practices—some languages do not separate with whitespace, while others do but there is no set practice." (p.9)

Strengths

- Text-only modality matches deployment exactly [\[Q3\]](#)
- Amharic Ethiopic script is included as a 'unique' script category [\[Q27\]](#); script diversity was an explicit design goal [\[Q20\]](#)
- Amharic uses space-delimited words, sidestepping the character-spacing issues that affected Japanese/Chinese/Khmer [\[Q80, Q92\]](#) — the 'default spacing provided by annotators' [\[Q82\]](#) applies

ID	Evidence
Q3	"With the English seed data included, there are 587k train utterances, 104k dev utterances, 152k test utterances, and 153k utterances currently held out for the MMNLU-22 competition, which will be released after the competition." (p.1)
Q20	"Fifth, we examined the script of each language, seeking to increase script diversity to drive experimentation in tokenization and normalization." (p.3)
Q27	"The Arabic script is used for three languages, the Cyrillic script for two languages, and the remaining 18 languages have "unique" scripts, in the sense that only one language in the dataset uses that script." (p.3)

Q80	“Three of these written languages, Japanese, Chinese (Traditional), and Chinese (Simplified), do not use spaces anywhere except to identify the end of a sentence. For these languages, we separate each character in the unlabeled input with a whitespace.” (p.7)
Q82	“We use the default spacing provided by annotators for all other languages.” (p.7)
Q92	“For Khmer, the workers had a difficult time adapting their translations and localizations to properly-slotted outputs given the space-optional nature of the language.” (p.8)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Both source and target are text in Ethiopic script; no signal-distribution mismatch on modality. Smartphone keyboard input methods (fidel, phonetic IME, occasional Latin-script Amharic) are not specifically validated by MASSIVE — INSUFFICIENT DOCUMENTATION on input-method robustness.

IF-2: Ethiopian smartphone infrastructure supports Ethiopic input; mobile penetration reached 61.4% Q1 2024 [WEB-9]; smartphone penetration ~15% nationally [WEB-8]. The deployment population is a self-selected app-user cohort with sufficient infrastructure.

IF-3: Amharic morphological richness (cliticization, root-and-pattern morphology — regional context) means slot boundaries may span morphologically complex tokens; MASSIVE acknowledges ‘collection system did not have a robust method to preserving or denoting native tokenization practices’ [Q109], leaving slot-boundary quality for Amharic unverified. amseg and HornMorpho tokenizers exist [WEB-17, WEB-19] but were not used.

IF-4: Form mismatches are minor: text-modality alignment is strong, but Amharic-specific tokenization quality and IME variants (fidel vs. romanized input) are not documented; this is a moderate but not severe external-validity risk.

ID	Evidence
Q1	“MASSIVE contains 1M realistic, parallel, labeled virtual assistant utterances spanning 51 languages, 18 domains, 60 intents, and 55 slots.” (p.1)
Q3	“With the English seed data included, there are 587k train utterances, 104k dev utterances, 152k test utterances, and 153k utterances currently held out for the MMNLU-22 competition, which will be released after the competition.” (p.1)
Q82	“We use the default spacing provided by annotators for all other languages.” (p.7)
Q109	“Fifth, our collection system did not have a robust method to preserving or denoting native tokenization practices—some languages do not separate with whitespace, while others do but there is no set practice.” (p.9)
WEB-8	Smartphone penetration ~15% nationally — deployment cohort is self-selected high-digital-literacy users (https://birmetrics.com/ethiopia-narrows-mobile-gender-gap-to-24-but-smartphone-access-for-women-remains-just-6/)
WEB-9	Mobile penetration 61.4% Q1 2024; 4G expanded to 45+ cities — adequate infrastructure for text-modality deployment (https://www.connectingafrica.com/connectivity/ethiopia-s-telecoms-liberalization-makes-progress-omdia)
WEB-17	amseg Amharic segmenter/tokenizer (Yimam et al. 2021) exists (https://github.com/EthioNLP/Ethiopian-Language-Survey)
WEB-19	HornMorpho rule-based morphological analyzer used in Ethiopian NLP pipelines (https://ethionlp.github.io/)

Information Gaps

- Whether MASSIVE Amharic utterances were tokenized with amseg or default whitespace
- Per-language slot-F1 sensitivity to tokenization choice for Amharic specifically

Requires Expert Verification

- Whether smartphone IME variation (fidel vs. phonetic vs. Latin) introduces real-world input distributions not represented in MASSIVE

Remediation

Gap 1: Amharic tokenization for slot boundaries not robustly handled by MASSIVE [Q109]; morphological richness may obscure slot spans

Recommendation: Re-tokenize Amharic training/eval data using amseg [WEB-17] or HornMorpho [WEB-19] morphological segmentation, and compare slot-F1 against MASSIVE's default whitespace tokenization to quantify the tokenization-quality effect on slot boundary detection.

ID	Evidence
Q109	"Fifth, our collection system did not have a robust method to preserving or denoting native tokenization practices—some languages do not separate with whitespace, while others do but there is no set practice." (p.9)
WEB-17	amseg Amharic segmenter/tokenizer (Yimam et al. 2021) exists (https://github.com/EthioNLP/Ethiopian-Language-Survey)
WEB-19	HornMorpho rule-based morphological analyzer used in Ethiopian NLP pipelines (https://ethionlp.github.io/)

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: medium

Benchmark: MASSIVE | Context: Ethiopian Intercity Travelers — Amharic-Language Booking Bot

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The output label space (60 intents and 55 slots **[Q1]**) is the same structure used to generate the input ontology, so the same gaps apply on the output side. Operator-specific slot types required by the deployment — operator_name, seat_class, route_closure_reason, surcharge_event, trip_change_type, departure_terminal — are absent from the 55-slot taxonomy (regional context, operator_specific_slot_types). The paper itself describes the schema as 'relatively simple when compared with hierarchical labeling schemata or flat schemata with more intent and slot options' **[Q108]**. Output decision rules (label/F1/exact match) are language-agnostic, so cultural decision-rule misalignment per se is limited — the violation is primarily content-validity (missing categories) rather than structural-validity for the labels that do exist. The text-to-text decoder format ('sequence of labels ... followed by the intent' **[Q73]**) is compatible with the deployment.

ID	Evidence
Q1	"MASSIVE contains 1M realistic, parallel, labeled virtual assistant utterances spanning 51 languages, 18 domains, 60 intents, and 55 slots." (p.1)
Q73	"The decoder output is a sequence of labels (including the Other label) for all of the tokens followed by the intent." (p.7)
Q108	"Fourth, our labeling schema is relatively simple when compared with hierarchical labeling schemata or flat schemata with more intent and slot options." (p.9)

Strengths

- 55-slot taxonomy includes generic temporal, location, and entity slots **[Q1]** that partially cover booking attributes
- Slot localization metadata (translate/localize/unchanged) is preserved **[Q58]**, allowing post-hoc filtering
- Judgment metadata for intent/slot semantic match is included **[Q63, Q64]**, supporting label-quality auditing

ID	Evidence
Q1	"MASSIVE contains 1M realistic, parallel, labeled virtual assistant utterances spanning 51 languages, 18 domains, 60 intents, and 55 slots." (p.1)
Q58	"The metadata of the released dataset includes whether the worker elected to "localize," "translate," or keep the slot "unchanged," primarily for the purposes of researchers evaluating machine translation systems, where it would be unreasonable to expect the system to "localize" to a specific song name the worker selected." (p.6)
Q63	"The output of the second workflow (the fully localized utterance) is judged by three workers for (1) whether the utterance matches the intent semantically, (2) whether the slots match their labels semantically, (3) grammaticality and naturalness, (4) spelling, and (5) language identification—English or mixed utterances are acceptable if that is natural for the language, but localizations without any tokens in the target language were not accepted." (p.6)
Q64	"These judgments are also included in the metadata of the dataset." (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
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OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

- OO-1:** Generic booking-relevant slot types (date, time, place_name, person, etc.) implied by 18 domains [Q1] partially apply. Ethiopian-specific slots (operator_name, seat_class, departure_terminal) are absent — confirmed by gap_id 11 (operator-specific seat-class taxonomy not publicly available).
- OO-2:** Missing categories: operator_name, seat_class, route_closure_reason, surcharge_event (holiday-driven), trip_change_type, departure_terminal — none are in the 55-slot taxonomy [Q1, Q108].
- OO-3:** The 60 intents reflect global virtual-assistant priorities [Q16] (smart-home, music, weather, etc.) that encode non-Ethiopian commercial assumptions; this is more construct-irrelevant variance than a values-encoding violation.
- OO-4:** Stakeholder-driven taxonomy redesign is needed for operator-specific slots; gap_id 11 confirms no public operator taxonomy exists, requiring direct elicitation.
- OO-5:** Structural validity is only mildly affected (label format is compatible); content validity is moderately to strongly affected by missing slot types [Q108].

ID	Evidence
Q1	“MASSIVE contains 1M realistic, parallel, labeled virtual assistant utterances spanning 51 languages, 18 domains, 60 intents, and 55 slots.” (p.1)
Q16	“Second, we determined existing languages available in major virtual assistants, such that the dataset could be used to benchmark today’s systems.” (p.3)
Q73	“The decoder output is a sequence of labels (including the Other label) for all of the tokens followed by the intent.” (p.7)
Q108	“Fourth, our labeling schema is relatively simple when compared with hierarchical labeling schemata or flat schemata with more intent and slot options.” (p.9)
WEB-21	No African transport-domain slot taxonomy exists — operator-specific slot types must be developed locally (https://aclanthology.org/2024.lrec-main.561/)

Information Gaps

- Detailed list of MASSIVE's 55 slot types to determine partial overlap with booking-domain needs

Requires Expert Verification

- Operator-specific seat-class enumeration and route-closure reason categories (gap_id 11)

Remediation

Gap 1: Operator-specific slot types (operator_name, seat_class, route_closure_reason, surcharge_event, trip_change_type, departure_terminal) missing from 55-slot taxonomy

Recommendation: Define an extended slot taxonomy with operator-specific types and an enumerated value set per slot (e.g., seat_class ∈ {VIP, economy, sleeper, ...} resolved via direct operator elicitation). Annotate the supplementary test set against this extended taxonomy, keeping MASSIVE-compatible slot types for backward comparability.

Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MASSIVE | Context: Ethiopian Intercity Travelers — Amharic-Language Booking Bot

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotators were vendor-pool MTurk workers selected via a listening-based fluency test [\[Q51\]](#), a three-judgment quality test [\[Q52\]](#), and a translator quiz with 'customized local-language examples' [\[Q53\]](#); workers self-certified as 'native speaker or ... fluent in the required language' [\[Q122\]](#). Annotator demographics — whether Amharic translators are urban Addis residents, Amhara-region speakers, or L2 speakers — are not disclosed (regional context, amharic_dialect_variation.massive_annotator_dialect_representation: NOT FOUND). The paper acknowledges 'the lack of labeled data ... that is realistic for the task and that is natural for each given language' [\[Q6\]](#) as the central problem the dataset attempts to address — but the translation-from-English pipeline means Amharic instances reflect English-language pragmatics translated into Amharic [\[Q2, Q4\]](#). The user has confirmed code-switching with Oromo (Q3 elicitation) will occur in real utterances, and gaps 3 and 4 confirm no Oromo–Amharic or Tigrinya–Amharic code-switching booking-domain corpus exists. 'Some low-quality utterances' are acknowledged [\[Q104\]](#) and removable only at the cost of further reducing the already small per-language sample [\[Q103, Q105\]](#). OC was rated HIGH priority and the violation is severe.

ID	Evidence
Q2	"MASSIVE was created by tasking professional translators to localize the English-only SLURP dataset into 50 typologically diverse languages from 29 genera." (p.1)
Q3	"With the English seed data included, there are 587k train utterances, 104k dev utterances, 152k test utterances, and 153k utterances currently held out for the MMNLU-22 competition, which will be released after the competition." (p.1)
Q4	"MASSIVE was created by localizing the SLURP NLU dataset (created only in English) in a parallel manner." (p.1)
Q6	"one difficulty in creating massively multilingual NLU models is the lack of labeled data for training and evaluation, particularly data that is realistic for the task and that is natural for each given language." (p.1)
Q51	"We offered two mechanisms to vendors for evaluating workers to be selected for each language. The first, which was used to select workers for the translation task, was an Amazon MTurk-hosted fluency test where workers listen to questions and statements in the relevant language and were evaluated using a multiple-choice questionnaire." (p.6)
Q52	"The second, which was used to select workers for the judgment task, was a test with a set of three judgments that the vendor could use to assess if workers were able to detect issues in the translated utterances." (p.6)
Q53	"In order to further improve worker selection quality, we created a translator quiz using the Amazon MTurk instructions that were created for translation and judgment tasks, coupled with customized local-language examples." (p.6)
Q103	"Starting with the dataset, the per-language data quantities are relatively small at 19.5k total records and 11.5k records for training." (p.9)
Q104	"Second, there are some low-quality utterances, both in the seed data and in the translations." (p.9)
Q105	"For the most part, these are surfaced through the judgment scores we provide for each record, but if a user does filtering based on these judgments, then the data size decreases even further." (p.9)
Q122	"By clicking 'SUBMIT', I also certify that I am a native speaker or am fluent in the required language" (p.17)

Strengths

- Multi-stage quality controls: fluency test [\[Q51\]](#), three-judgment QA [\[Q52\]](#), translator quiz [\[Q53\]](#), MT-detection alarms [\[Q66\]](#)
- Three independent judgment scores per utterance covering intent match, slot match, grammaticality, spelling, language identification [\[Q63\]](#) — included in metadata [\[Q64\]](#) for filtering

- Self-certified native/fluent speakers [Q122] and vendor monitoring for engagement quality [Q44, Q66]

ID	Evidence
Q44	"This vendor pool was reduced to three based on engagement and resource availability." (p.5)
Q51	"We offered two mechanisms to vendors for evaluating workers to be selected for each language. The first, which was used to select workers for the translation task, was an Amazon MTurk-hosted fluency test where workers listen to questions and statements in the relevant language and were evaluated using a multiple-choice questionnaire." (p.6)
Q52	"The second, which was used to select workers for the judgment task, was a test with a set of three judgments that the vendor could use to assess if workers were able to detect issues in the translated utterances." (p.6)
Q53	"In order to further improve worker selection quality, we created a translator quiz using the Amazon MTurk instructions that were created for translation and judgment tasks, coupled with customized local-language examples." (p.6)
Q63	"The output of the second workflow (the fully localized utterance) is judged by three workers for (1) whether the utterance matches the intent semantically, (2) whether the slots match their labels semantically, (3) grammaticality and naturalness, (4) spelling, and (5) language identification—English or mixed utterances are acceptable if that is natural for the language, but localizations without any tokens in the target language were not accepted." (p.6)
Q64	"These judgments are also included in the metadata of the dataset." (p.6)
Q66	"Workers were also monitored to see if their tasks were primarily machine translated. Such workers were removed from the pool and all of their work was resubmitted to be completed by the other workers." (p.6)
Q122	"By clicking 'SUBMIT', I also certify that I am a native speaker or am fluent in the required language" (p.17)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: No — labels reflect English-source intent/slot judgments translated into Amharic [Q2, Q4]; they do not reflect Ethiopian intercity-traveler stakeholder perspectives. Code-switching, L2-speaker phrasing, and operator-specific slot judgments are not represented (gaps 3, 4, 5).

OC-2: Substantial disagreement is likely between MASSIVE Amharic annotators (translators of English seed data) and the actual Ethiopian intercity-traveler population, particularly for code-switched and Ethiopian-calendar-bearing utterances (Q3 elicitation; gaps 3, 4, 5).

OC-3: Demographic information for Amharic translators is not disclosed in the paper — workers are described only as fluency-tested vendors [Q41, Q42, Q51, Q122]. The regional context confirms 'NOT FOUND' for translator regional/dialect background.

OC-4: Re-annotation by representative Ethiopian intercity travelers (urban Addis, regional Amhara, Amharic-as-L2 speakers) is needed; gap_id 5 confirms no existing L2-speaker NLP dataset to bootstrap from.

OC-5: Aggregation uses three judgment workers per utterance [Q63]; if the three-worker pool is demographically homogeneous (e.g., all urban Addis fluent speakers), L2-speaker and code-switched perspectives are systematically excluded. INSUFFICIENT DOCUMENTATION on judgment-worker demographics.

OC-6: Severe convergent and external-validity risks: labels do not correlate with regional stakeholder judgments for the linguistic phenomena (code-switching, L2 phrasing, calendar/operator slots) that the deployment must handle [Q6, Q104,

Q107].

ID	Evidence
Q2	"MASSIVE was created by tasking professional translators to localize the English-only SLURP dataset into 50 typologically diverse languages from 29 genera." (p.1)
Q3	"With the English seed data included, there are 587k train utterances, 104k dev utterances, 152k test utterances, and 153k utterances currently held out for the MMNLU-22 competition, which will be released after the competition." (p.1)
Q4	"MASSIVE was created by localizing the SLURP NLU dataset (created only in English) in a parallel manner." (p.1)
Q6	"one difficulty in creating massively multilingual NLU models is the lack of labeled data for training and evaluation, particularly data that is realistic for the task and that is natural for each given language." (p.1)
Q41	"The MASSIVE dataset was collected using a customized workflow powered by Amazon MTurk." (p.5)
Q42	"We required a vendor pool with the capability and resources to collect a large multilingual dataset." (p.5)
Q51	"We offered two mechanisms to vendors for evaluating workers to be selected for each language. The first, which was used to select workers for the translation task, was an Amazon MTurk-hosted fluency test where workers listen to questions and statements in the relevant language and were evaluated using a multiple-choice questionnaire." (p.6)
Q63	"The output of the second workflow (the fully localized utterance) is judged by three workers for (1) whether the utterance matches the intent semantically, (2) whether the slots match their labels semantically, (3) grammaticality and naturalness, (4) spelling, and (5) language identification—English or mixed utterances are acceptable if that is natural for the language, but localizations without any tokens in the target language were not accepted." (p.6)
Q104	"Second, there are some low-quality utterances, both in the seed data and in the translations." (p.9)
Q107	"Relatedly, allowing the worker to decide on translation versus localization of slot entities added further noise to the dataset, although we try to store this decision in the metadata." (p.9)
Q122	"By clicking "SUBMIT", I also certify that I am a native speaker or am fluent in the required language" (p.17)
WEB-1	Oromo speakers >40% of Ethiopia's population — large L2-Amharic share among intercity travelers (https://arxiv.org/html/2403.19365v1)
WEB-18	Amharic–Afaan Oromo bilingual code-switching corpus exists for hate speech only — domain mismatch with booking (https://link.springer.com/article/10.1186/s40537-024-01044-y)
WEB-20	Amharic/Oromo/Tigrigna code-switching empirically documented in Ethiopian social media — confirms the phenomenon is real and is unrepresented in monolingual MASSIVE Amharic (https://aclanthology.org/2024.rail-1.13.pdf)

Information Gaps

- Vendor identity and translator regional/dialect demographics for Amharic
- Distribution of judgment scores for Amharic utterances and how filtering by score affects size

Requires Expert Verification

- Empirical disagreement rate between MASSIVE Amharic labels and Ethiopian intercity-traveler relabels on a sample
- Whether code-switched Amharic–Oromo utterances would be rejected by MASSIVE's language-identification judgment [Q63]

Remediation

Gap 1: Amharic translator demographics undisclosed; no representation of L2-Amharic speakers or code-switching with Oromo/Tigrinya, despite user confirmation that these occur in real utterances

Recommendation: Recruit a regional annotator pool stratified across urban Addis, Amhara region, and L2-Amharic speakers from Oromia and Tigray corridors. Re-label a sample of MASSIVE Amharic utterances and measure inter-annotator agreement against original labels; then build a code-switched evaluation set drawing on the Amharic–Oromo corpus [WEB-18] and RAIL 2024 lexicon [WEB-20] adapted to booking-domain vocabulary.

ID	Evidence
WEB-18	Amharic–Afaan Oromo bilingual code-switching corpus exists for hate speech only — domain mismatch with booking (https://link.springer.com/article/10.1186/s40537-024-01044-y)

WEB-20

Amharic/Oromo/Tigrigna code-switching empirically documented in Ethiopian social media — confirms the phenomenon is real and is unrepresented in monolingual MASSIVE Amharic (<https://aclanthology.org/2024.rail-1.13.pdf>)

Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: MASSIVE | Context: Ethiopian Intercity Travelers — Amharic-Language Booking Bot

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MASSIVE's output format — intent classification + token-level slot-fill sequence [Q70, Q71, Q73] — directly matches what an intent-and-slot booking system requires. Three primary metrics are reported: 'exact match accuracy, intent classification accuracy, and slot-filling F1 score' [Q5], with per-language breakdowns [Q126, Q127, Q128] and 95% confidence intervals [Q89]. Per-language Amharic results are reported, allowing direct estimation of expected performance. However, two structural concerns remain: (1) Pearson correlation of 0.54 between XLM-R pretraining data quantity and zero-shot exact match [Q87], combined with the finding that 'Germanic genera and Latin scripts performed the best overall' [Q98] and that low-pretraining-data languages like Icelandic underperformed [Q100], strongly suggests Amharic — a unique-script, lower-resource language — will fall in the lower-performance tier; (2) deployment may eventually require text-to-speech for accessibility (literacy ~51.8% in 2017, urban–rural gap [WEB-3, WEB-12, WEB-11]), but MASSIVE evaluates only text outputs [Q5]. OF was rated LOWER priority and overall format alignment is strong.

ID	Evidence
Q5	"We also present modeling results on XLM-R and mT5, including exact match accuracy, intent classification accuracy, and slot-filling F1 score." (p.1)
Q70	"The first classification head used the pooled output from the encoder to predict the intent and the second used the sequence output to predict the slots." (p.7)
Q71	"As pooling for the intent classification head, we experimented with using hidden states from the first position, averaged hidden states across the sequence, and the maximally large hidden state from the sequence." (p.7)
Q73	"The decoder output is a sequence of labels (including the Other label) for all of the tokens followed by the intent." (p.7)
Q87	"We compared the pretraining data quantities by language for XLM-R to its per-language task performance values, and in the zero shot setup, we found a Pearson correlation of 0.54 for exact match" (p.7)
Q89	"Intervals for 95% confidence are given assuming normal distributions." (p.8)
Q98	"Discounting for artificial spacing effects, Germanic genera and Latin scripts performed the best overall (See Appendix E), which is unsurprising given the amount of pretraining data for those genera and scripts, as well as the quantity of Germanic and Latin-script languages in MASSIVE." (p.8)
Q100	"Icelandic was the lowest-performing Germanic language, likely due to a lack of pretraining data, as well as to its linguistic evolution away from the others due to isolated conditions." (p.8)
Q126	"Exact match accuracy by language for our three models using the full dataset and the zero-shot setup." (p.20)
Q127	"Table 8: Intent accuracy by language for our three models using the full dataset and the zero-shot setup." (p.21)
Q128	"Micro-averaged slot-filling F1 by language for our three models using the full dataset and the zero-shot setup." (p.22)
WEB-3	Ethiopia adult literacy ~51.8% (2017) (https://www.cia.gov/the-world-factbook/about/archives/2023/countries/ethiopia/)
WEB-11	Significant urban–rural literacy gap; pastoralist regions much lower (https://zoetalentsolutions.com/education-statistics-for-ethiopia/)
WEB-12	UNESCO literacy data corroborates ~51.8% figure (https://www.theglobaleconomy.com/Ethiopia/Literacy_rate/)

Strengths

- Output format (intent label + slot-token sequence) directly matches deployment requirement [Q70, Q73]
- Three primary metrics (exact match, intent accuracy, slot F1) match standard NLU evaluation [Q5]

- Per-language results published for Amharic [Q126, Q127, Q128] allowing direct expected-performance estimation
- Reports both full-dataset and zero-shot performance [Q83, Q85] enabling lower-bound estimation for low-resource transfer scenarios

ID	Evidence
Q5	"We also present modeling results on XLM-R and mT5, including exact match accuracy, intent classification accuracy, and slot-filling F1 score." (p.1)
Q70	"The first classification head used the pooled output from the encoder to predict the intent and the second used the sequence output to predict the slots." (p.7)
Q73	"The decoder output is a sequence of labels (including the Other label) for all of the tokens followed by the intent." (p.7)
Q83	"Zero-shot performance was also assessed, in which the models were trained on English data, validation was performed on all languages, and testing was performed on all non-English locales." (p.7)
Q85	"Zero-shot exact match performance is 25-37 points worse than that of full-dataset training runs." (p.7)
Q126	"Exact match accuracy by language for our three models using the full dataset and the zero-shot setup." (p.20)
Q127	"Table 8: Intent accuracy by language for our three models using the full dataset and the zero-shot setup." (p.21)
Q128	"Micro-averaged slot-filling F1 by language for our three models using the full dataset and the zero-shot setup." (p.22)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Yes — text-based intent + slot output matches the booking-bot deployment exactly [Q70, Q71, Q73].

OF-2: MASSIVE evaluates text only [Q5]; if accessibility for lower-literacy users requires TTS output (literacy ~51.8% adult, [WEB-3, WEB-12]; rural literacy lower [WEB-11]), MASSIVE provides no signal on TTS quality. INSUFFICIENT DOCUMENTATION on speech-output evaluation.

OF-3: Adult literacy ~51.8% (2017 data, [WEB-3, WEB-12]); the deployment's smartphone-app cohort is a higher-literacy self-selected subset, but accessibility for less literate or older travelers may require speech output not evaluated by MASSIVE.

OF-4: Form mismatches are minor for the text-NLU output but expected Amharic performance is in the lower tier per the script/pretraining-data correlation [Q87, Q98, Q100]; this is a performance-distribution concern rather than a form mismatch.

ID	Evidence
Q5	"We also present modeling results on XLM-R and mT5, including exact match accuracy, intent classification accuracy, and slot-filling F1 score." (p.1)
Q70	"The first classification head used the pooled output from the encoder to predict the intent and the second used the sequence output to predict the slots." (p.7)
Q71	"As pooling for the intent classification head, we experimented with using hidden states from the first position, averaged hidden states across the sequence, and the maximally large hidden state from the sequence." (p.7)
Q73	"The decoder output is a sequence of labels (including the Other label) for all of the tokens followed by the intent." (p.7)
Q87	"We compared the pretraining data quantities by language for XLM-R to its per-language task performance values, and in the zero shot setup, we found a Pearson correlation of 0.54 for exact match" (p.7)
Q98	"Discounting for artificial spacing effects, Germanic genera and Latin scripts performed the best overall (See Appendix E), which is unsurprising given the amount of pretraining data for those genera and scripts, as well as the quantity of Germanic and Latin-script languages in MASSIVE." (p.8)

Q100	"Icelandic was the lowest-performing Germanic language, likely due to a lack of pretraining data, as well as to its linguistic evolution away from the others due to isolated conditions." (p.8)
WEB-3	Ethiopia adult literacy ~51.8% (2017) (https://www.cia.gov/the-world-factbook/about/archives/2023/countries/ethiopia/)
WEB-11	Significant urban–rural literacy gap; pastoralist regions much lower (https://zoetalentsolutions.com/education-statistics-for-ethiopia/)
WEB-12	UNESCO literacy data corroborates ~51.8% figure (https://www.theglobaleconomy.com/Ethiopia/Literacy_rate/)

Information Gaps

- Reported MASSIVE Amharic exact-match / intent-acc / slot-F1 numbers (Tables 7–9, [Q126, Q127, Q128]) were not surfaced in the documentation extract

Requires Expert Verification

- Whether the deployment ultimately requires TTS output for accessibility, beyond text-only NLU

Remediation

Gap 1: Expected lower-tier performance for Amharic per pretraining-data and script correlations [Q87, Q98, Q100]; reliance on text-only output may underserve lower-literacy users (~51.8% adult literacy)

Recommendation: Report Amharic-specific exact-match, intent-accuracy, and slot-F1 separately rather than aggregated; set deployment thresholds based on Amharic-only metrics. If accessibility for lower-literacy users is in scope, supplement with a TTS-output evaluation track outside MASSIVE.

ID	Evidence
Q87	"We compared the pretraining data quantities by language for XLM-R to its per-language task performance values, and in the zero shot setup, we found a Pearson correlation of 0.54 for exact match" (p.7)
Q98	"Discounting for artificial spacing effects, Germanic genera and Latin scripts performed the best overall (See Appendix E), which is unsurprising given the amount of pretraining data for those genera and scripts, as well as the quantity of Germanic and Latin-script languages in MASSIVE." (p.8)
Q100	"Icelandic was the lowest-performing Germanic language, likely due to a lack of pretraining data, as well as to its linguistic evolution away from the others due to isolated conditions." (p.8)